



دانشگاه صنعتی امیرکبیر

Software Requirements Specification (SRS)

End-to-End Cinema Ticketing Platform

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Document Information

Item	Description
Project Name	Cinema Ticketing Platform
Course	Software Engineering II
Document Version	1.0
Target UI Language	Persian / Dari
Text Direction	Right-to-Left, RTL
System Type	Web-based cinema ticketing platform
Main Purpose	To build a realistic cinema ticketing web application with role-based access, movie management, showtime management, seat selection, temporary seat locking, mock payment, QR ticket generation, and staff ticket validation

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional and non-functional requirements of the Cinema Ticketing Platform.

The main purpose of the system is to provide a realistic web-based cinema ticketing solution where customers can browse available movies, select showtimes, choose seats from a visual seat map, complete a mock payment, receive QR-code tickets, and enter the cinema after staff validation.

The system must also allow cinema managers to manage movies, cinemas, halls, seat layouts, showtimes, ticket prices, and sales reports. In addition, staff members must be able to validate tickets at the cinema entrance.

1.2 Project Scope

The system will be developed as a Persian-first and RTL-supported cinema ticketing platform. It must support real user roles, real booking flow, seat states, temporary seat locking, and realistic dashboards.

The platform should not be a simple landing page. It must behave like a usable product that covers the full cinema ticket purchasing process from movie discovery to ticket validation.

The main scope of the project includes:

1. User authentication and role-based access.
2. Movie and cinema management.
3. Showtime and hall management.
4. Visual seat selection.
5. Temporary seat locking.
6. Double-booking prevention.
7. Mock payment processing.
8. Ticket generation with QR data.
9. Staff ticket validation.
10. Admin-level platform monitoring and management.

1.3 Target Users

The system has four main user roles:

Role	Main Responsibility
Customer	Browse movies, select showtimes, reserve seats, complete payment, and receive tickets
Cinema Manager	Manage movies, cinemas, halls, seat layouts, showtimes, ticket prices, and sales reports
Staff	Validate tickets at the cinema entrance and mark valid tickets as used
Admin	Manage users, cinemas, roles, platform statistics, and system status

2. Product Vision

The Cinema Ticketing Platform is a realistic web application for online cinema ticket sales.

A customer should be able to enter the website, log in, browse available movies, select a cinema and showtime, choose seats from a visual seat map, reserve seats temporarily, complete a mock payment, and receive a final ticket with a unique ticket code and QR data.

A cinema manager should be able to add movies, define cinema halls, create seat layouts, schedule showtimes, set ticket prices, and view sales reports.

A staff member should be able to validate tickets by using the ticket code or QR data. If the ticket is valid, the system should mark it as used. If the ticket is invalid, cancelled, or already used, the system should reject it with a clear message.

An admin should be able to monitor the entire platform, manage users, manage cinemas, control roles, and view system statistics.

3. System Objectives

The main objectives of the system are:

1. Provide a Persian/Dari-first cinema ticketing experience.
2. Support full RTL layout.
3. Support role-based login and dashboards.
4. Allow cinema managers to create and manage real cinema content.
5. Allow customers to browse movies and buy tickets.
6. Prevent double-booking of cinema seats.
7. Temporarily lock selected seats during checkout.
8. Generate final tickets after successful payment.
9. Allow staff members to validate tickets at the cinema entrance.

10. Provide admin-level platform management.

11. Keep the system clean, scalable, and suitable for a Software Engineering II final project.

4. User Roles and Permissions

4.1 Customer

A customer can perform the following actions:

Feature	Description
Register / Login	Create an account and log in to the system
Browse Movies	View published movies
Search and Filter	Search movies by title, city, cinema, genre, and date
View Movie Details	View movie information and available showtimes
Select Showtime	Choose a cinema, hall, and showtime
Select Seats	Select one or more available seats
Lock Seats	Temporarily lock selected seats before payment
Pay	Complete mock payment
Receive Ticket	Receive a ticket code and QR data
My Tickets	View purchased tickets

4.2 Cinema Manager

A cinema manager can perform the following actions:

Feature	Description
Manager Dashboard	View cinema-level summary
Add Movie	Create new movie records
Edit Movie	Update movie information
Add Poster	Add poster URL for a movie
Manage Cinema	Add or edit cinema information
Manage Halls	Create halls and define capacity
Manage Seats	Generate seat layout for halls
Create Showtimes	Schedule movie showtimes
Set Price	Define ticket price per showtime
Sales Report	View sales and remaining capacity

4.3 Staff

A staff member can perform the following actions:

Feature	Description
Staff Dashboard	View staff tools
Search Ticket	Search for a ticket by ticket code
Validate Ticket	Validate a ticket code or QR data
Mark as Used	Mark a valid ticket as used
Reject Invalid Ticket	Reject fake or unknown tickets
Reject Used Ticket	Reject tickets that have already been used

4.4 Admin

An admin can perform the following actions:

Feature	Description
Admin Dashboard	View platform summary
Manage Users	View and manage user accounts
Manage Roles	Control role assignments
Manage Cinemas	View and manage cinemas
View Statistics	View movies, showtimes, payments, tickets, and revenue statistics
System Status	View system health and monitoring information

5. Functional Requirements

5.1 Authentication and Authorization

ID	Requirement
FR-001	The system shall allow users to log in using email and password.
FR-002	The system shall support role-based access control.
FR-003	The system shall display different dashboards based on the user role.
FR-004	The system shall prevent users from accessing pages outside their role.
FR-005	The system shall provide demo accounts for testing.

Demo Accounts

Role	Email	Password
Admin	admin@example.com	password123
Cinema Manager	manager@example.com	password123
Staff	staff@example.com	password123
Customer	customer@example.com	password123

5.2 Movie Management

ID	Requirement
FR-006	Cinema managers shall be able to create movies.
FR-007	Cinema managers shall be able to edit movies.
FR-008	Each movie shall have a title, description, genre, duration, language, age rating, poster URL, and status.
FR-009	Movies shall have either Draft or Published status.
FR-010	Only published movies shall be visible to customers.

Movie Fields

Field	Description
Title	Movie name
Description	Short explanation about the movie
Genre	Movie category such as Drama, Action, Comedy, or Family
Duration	Duration of the movie in minutes
Language	Movie language
Age Rating	Age category of the movie
Poster URL	Image URL for the movie poster
Status	Movie status: Draft or Published

5.3 Cinema, Hall, and Seat Management

ID	Requirement
FR-011	Cinema managers shall be able to add cinema information.
FR-012	A cinema shall have a name, city, address, and phone number.
FR-013	Cinema managers shall be able to create halls.
FR-014	A hall shall have a name and capacity.
FR-015	The system shall generate seats based on rows and seats per row.
FR-016	Each seat shall have a readable label such as A1, A2, or B1.

5.4 Showtime Management

ID	Requirement
FR-017	Cinema managers shall be able to create showtimes.
FR-018	Each showtime shall belong to one movie, one cinema, and one hall.
FR-019	Each showtime shall have start time, end time, ticket price, and status.
FR-020	Published showtimes shall be visible to customers.
FR-021	The system shall show the number of remaining seats for each showtime.

5.5 Movie Discovery

ID	Requirement
FR-022	Customers shall be able to view all published movies.
FR-023	Customers shall be able to search movies by title.
FR-024	Customers shall be able to filter movies by city, cinema, genre, and date.
FR-025	Customers shall be able to open the movie details page.
FR-026	The movie details page shall show available showtimes.

5.6 Seat Selection and Seat Locking

ID	Requirement
FR-027	Customers shall be able to view a visual seat map.
FR-028	Seats shall show one of the following states: Available, Selected, Locked, or Booked.
FR-029	Customers shall be able to select one or more seats.
FR-030	The system shall temporarily lock selected seats before payment.
FR-031	The seat lock shall expire after a fixed time, preferably 10 minutes.
FR-032	The UI shall show a reservation countdown timer.
FR-033	Locked seats shall not be selectable by other users.
FR-034	Seats shall be released if payment fails, reservation expires, or the customer cancels the reservation.

Recommended Redis Lock Key

`seat_lock:{showtimeId}:{seatId}`

5.7 Double-Booking Prevention

ID	Requirement
FR-035	The system shall prevent two customers from buying the same seat for the same showtime.
FR-036	Temporary Redis seat locking shall prevent concurrent seat selection conflicts.
FR-037	Database-level uniqueness shall prevent final duplicate tickets.
FR-038	A ticket shall be unique by showtime and seat.

Recommended Database Rule

`UNIQUE(showtime_id, seat_id)`

5.8 Checkout and Payment

ID	Requirement
FR-039	Customers shall see the reservation summary before payment.
FR-040	The checkout page shall show movie, cinema, hall, seats, showtime, and total price.

ID	Requirement
FR-041	The system shall support mock payment success.
FR-042	The system shall support mock payment failure.
FR-043	Successful payment shall confirm the reservation.
FR-044	Failed payment shall cancel the reservation and release the seats.
FR-045	A ticket shall only be issued after successful payment.

5.9 Ticket Issuance

ID	Requirement
FR-046	The system shall generate a ticket after successful payment.
FR-047	Each ticket shall have a unique ticket code.
FR-048	The ticket code shall be readable and human-friendly.
FR-049	Each ticket shall include QR data.
FR-050	Customers shall be able to view their tickets.

Recommended Ticket Code Format

CIN-YYYYMMDD-XXXXXX

Ticket Details

Field	Description
Ticket Code	Unique readable ticket code
Movie	Movie title
Cinema	Cinema name
Hall	Hall name
Seat	Seat label
Showtime	Date and time of the showtime
Status	Ticket status
QR Code	QR data or QR image

5.10 Ticket Validation

ID	Requirement
FR-051	Staff shall be able to search for a ticket by ticket code.
FR-052	Staff shall be able to validate tickets.
FR-053	Valid tickets shall be marked as Used.
FR-054	Already used tickets shall be rejected.
FR-055	Cancelled tickets shall be rejected.
FR-056	Invalid ticket codes shall be rejected.
FR-057	The system shall show clear validation messages.

Ticket Statuses

Status	Meaning
VALID	The ticket is valid and can be used
USED	The ticket has already been used
CANCELLED	The ticket has been cancelled

5.11 Notifications

ID	Requirement
FR-058	The system shall create notification records for important events.
FR-059	Events shall include reservation, payment, ticket issuance, and ticket validation.
FR-060	SMS and email can be simulated in the MVP version.
FR-061	The architecture shall support future SMS and email integration.

5.12 Admin Dashboard

ID	Requirement
FR-062	Admin shall view platform summary.
FR-063	Admin shall view total users, cinemas, movies, showtimes, reservations, payments, tickets, and revenue.
FR-064	Admin shall manage users.
FR-065	Admin shall manage cinemas.
FR-066	Admin shall view system health and monitoring information.

6. Non-Functional Requirements

6.1 Usability

ID	Requirement
NFR-001	The UI shall be Persian/Dari-first.
NFR-002	The UI shall fully support RTL layout.
NFR-003	The UI shall be simple and clear for non-technical users.
NFR-004	Buttons, messages, labels, and menus shall be understandable.

6.2 Performance

ID	Requirement
NFR-005	Movie and showtime pages should load quickly.
NFR-006	Seat locking should respond quickly.
NFR-007	The system should handle concurrent users trying to book seats.

6.3 Reliability

ID	Requirement
NFR-008	Confirmed tickets must not be duplicated.
NFR-009	Failed payments must not generate tickets.
NFR-010	Expired reservations must not block seats forever.

6.4 Security

ID	Requirement
NFR-011	Passwords shall be stored securely.
NFR-012	Role-based pages shall be protected.
NFR-013	Customers shall not access other customers' tickets.
NFR-014	Staff shall not modify movie, payment, or admin data.

6.5 Maintainability

ID	Requirement
NFR-015	The project shall use a clean structure.
NFR-016	Components and pages shall be readable and organized.
NFR-017	Database relationships shall be meaningful and normalized.
NFR-018	The project shall include setup and usage instructions.

7. Data Model Requirements

7.1 Main Entities

The system shall include the following main entities:

Entity	Purpose
User	Stores system users and roles
Cinema	Stores cinema information
Hall	Stores cinema halls
Seat	Stores hall seats
Movie	Stores movie information
Showtime	Stores movie schedules
Reservation	Stores temporary or confirmed reservations
Reservation Seat	Stores selected seats for a reservation
Payment	Stores payment transactions
Ticket	Stores issued tickets
Notification	Stores system notifications

7.2 Entity Fields

User

Field	Description
id	Unique user ID
full_name	User full name
email	User email
password_hash	Hashed password
role	User role
created_at	Account creation date

Cinema

Field	Description
id	Unique cinema ID
name	Cinema name
city	Cinema city
address	Cinema address
phone	Cinema phone number
manager_id	Related cinema manager
created_at	Creation date

Hall

Field	Description
id	Unique hall ID
cinema_id	Related cinema
name	Hall name
capacity	Total number of seats
created_at	Creation date

Seat

Field	Description
id	Unique seat ID
hall_id	Related hall
row_label	Row label
seat_number	Seat number
seat_label	Human-readable seat label

Movie

Field	Description
id	Unique movie ID
title	Movie title
description	Movie description
genre	Movie genre
duration_minutes	Movie duration in minutes
language	Movie language
age_rating	Movie age rating
poster_url	Poster image URL
status	Movie status: Draft or Published
created_by	Creator manager
created_at	Creation date
updated_at	Last update date

Showtime

Field	Description
id	Unique showtime ID
movie_id	Related movie
cinema_id	Related cinema
hall_id	Related hall
start_time	Showtime start date and time
end_time	Showtime end date and time
ticket_price	Ticket price
status	Showtime status: Draft or Published

Reservation

Field	Description
id	Unique reservation ID
user_id	Related customer
showtime_id	Related showtime
status	Reservation status: Pending, Confirmed, Cancelled, or Expired
total_amount	Total reservation amount
expires_at	Seat lock expiration time
created_at	Creation date

Payment

Field	Description
id	Unique payment ID
reservation_id	Related reservation

Field	Description
user_id	Related customer
amount	Payment amount
status	Payment status: Pending, Successful, or Failed
provider	Mock payment provider
created_at	Creation date

Ticket

Field	Description
id	Unique ticket ID
ticket_code	Human-readable ticket code
user_id	Related customer
showtime_id	Related showtime
seat_id	Related seat
payment_id	Related payment
status	Ticket status: Valid, Used, or Cancelled
qr_data	QR information
issued_at	Ticket issue date
used_at	Ticket validation date

8. Core Workflows

8.1 Customer Ticket Purchase Flow

1. Customer opens the website.
2. Customer logs in.
3. Customer views the movie list.
4. Customer searches or filters movies.
5. Customer opens the movie details page.
6. Customer selects a showtime.
7. Customer views the seat map.
8. Customer selects one or more seats.
9. The system temporarily locks the selected seats.
10. Customer sees the reservation countdown timer.
11. Customer proceeds to checkout.
12. Customer completes mock payment.
13. The system confirms the reservation.

14.The system generates the final ticket.

15.Customer views the ticket with ticket code and QR data.

8.2 Cinema Manager Flow

1. Manager logs in.
2. Manager opens the manager dashboard.
3. Manager adds or edits cinema information.
4. Manager creates halls and seats.
5. Manager adds movies.
6. Manager creates showtimes.
7. Published showtimes become visible to customers.
8. Manager views the sales report.

8.3 Staff Ticket Validation Flow

1. Staff logs in.
2. Staff opens the ticket validation page.
3. Customer presents a ticket code or QR code.
4. Staff searches for the ticket code.
5. The system checks the ticket status.
6. If the ticket is valid, the system marks it as used.
7. If the ticket is invalid, used, or cancelled, the system rejects the ticket.

8.4 Admin Flow

1. Admin logs in.
2. Admin opens the admin dashboard.
3. Admin views platform statistics.
4. Admin manages users and cinemas.
5. Admin checks system health and monitoring information.

9. UI/UX Requirements

9.1 General UI Requirements

The application UI shall be:

1. Persian/Dari-first.
2. Fully RTL-supported.
3. Clean and modern.
4. Responsive on different screen sizes.
5. Easy to understand.
6. Suitable for a real cinema ticketing product.

9.2 Recommended Navigation

Menu Item	Purpose
Home	Opens the main page
Movies	Shows available movies
My Tickets	Shows purchased tickets
Manager Dashboard	Opens the cinema manager panel
Staff Validation	Opens the ticket validation page
Admin Dashboard	Opens the admin panel
Logout	Logs the user out

9.3 Required Pages

Customer Pages

1. Login page
2. Home page
3. Movies page
4. Movie details page
5. Showtime selection page
6. Seat selection page
7. Checkout page
8. Final ticket page

9. My tickets page

Manager Pages

1. Manager dashboard
2. Add movie page
3. Manage movies page
4. Add cinema page
5. Add hall page
6. Create showtime page
7. Sales report page

Staff Pages

1. Staff dashboard
2. Ticket search page
3. Ticket validation page

Admin Pages

1. Admin dashboard
2. User management page
3. Cinema management page
4. Platform statistics page
5. System status page

10. Suggested API Structure

10.1 Auth APIs

Method	Endpoint	Purpose
POST	/api/auth/register	Register a user
POST	/api/auth/login	Log in a user
GET	/api/auth/me	Get current user information

10.2 Movie APIs

Method	Endpoint	Purpose
GET	/api/movies	List movies
GET	/api/movies/:id	Get movie details
GET	/api/movies/:id/showtimes	Get showtimes for a movie

10.3 Showtime APIs

Method	Endpoint	Purpose
GET	/api/showtimes	List showtimes
GET	/api/showtimes/:id	Get showtime details
GET	/api/showtimes/:id/seat-map	Get seat map for a showtime

10.4 Reservation APIs

Method	Endpoint	Purpose
POST	/api/reservations/lock-seat	Lock selected seats
GET	/api/reservations/:id	Get reservation details
POST	/api/reservations/:id/cancel	Cancel reservation

10.5 Payment APIs

Method	Endpoint	Purpose
POST	/api/checkout	Create checkout
POST	/api/payments/mock-success/:id	Simulate successful payment
POST	/api/payments/mock-fail/:id	Simulate failed payment

10.6 Ticket APIs

Method	Endpoint	Purpose
GET	/api/tickets/my/:userId	Get customer tickets
GET	/api/tickets/:id	Get ticket details
POST	/api/staff/tickets/validate-code	Validate ticket by code

10.7 Manager APIs

Method	Endpoint	Purpose
GET	/api/manager/dashboard	Get manager dashboard summary
POST	/api/manager/movies	Create movie
POST	/api/manager/cinemas	Create cinema
POST	/api/manager/halls	Create hall
POST	/api/manager/showtimes	Create showtime
GET	/api/manager/sales-report	Get sales report

10.8 Admin APIs

Method	Endpoint	Purpose
GET	/api/admin/dashboard	Get admin dashboard summary
GET	/api/admin/users	Get user list
GET	/api/admin/cinemas	Get cinema list
GET	/api/admin/system-summary	Get system status summary

11. Acceptance Criteria

The project will be acceptable when the following criteria are met:

ID	Acceptance Criterion
AC-001	Customer can log in.
AC-002	Customer can browse movies.
AC-003	Customer can select a showtime.
AC-004	Customer can select seats from a seat map.
AC-005	The system temporarily locks selected seats.
AC-006	Two users cannot buy the same seat for the same showtime.
AC-007	Customer can complete mock payment.
AC-008	A ticket is generated after successful payment.
AC-009	Customer can view the ticket with code and QR data.
AC-010	Staff can validate a ticket.
AC-011	Staff cannot validate the same ticket twice.
AC-012	Manager can create movies, halls, and showtimes.
AC-013	Manager-created showtimes appear for customers.
AC-014	Admin can view platform statistics.
AC-015	UI is Persian/Dari and RTL-supported.
AC-016	Project has clean structure and setup instructions.

12. Conclusion

The Cinema Ticketing Platform must work like a real ticketing product. It should allow customers to complete the full ticket purchase process from movie discovery to final ticket generation.

The most important parts of the system are role-based access, movie and showtime management, visual seat selection, temporary seat locking, double-booking prevention, mock payment, QR ticket generation, and staff ticket validation.

The system should also include manager and admin dashboards so that the project is complete from the perspective of software design, architecture, data modeling, user roles, and workflow management.